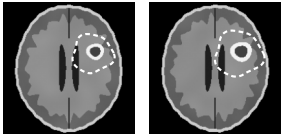
A At each follow up MRI after treatment, has the tumour progressed?

PRISM keeps image and clinical-history evidence apart and reports them side by side

1 Inputs
MRI, released masks, record



previous current

T1-contrast and FLAIR MRI

surgery scan

chemoRT

age · sex · IDH · MGMT

treatment dates, clinical record

2 Measure
27 measurements per scan

Segmentation
enhancing tumour
4.0 → 6.0 cc
FLAIR 31.0 cc

Position in treatment
8 weeks after the end of radiotherapy, with the clinical record

3 Score
3 ridge logistic models

Imaging 0.64

Combined 0.33

Clinical history 0.21

example values (Figure 5)

4 Confidence
split conformal, 80% level

80% conformal set
{progression, no progression}
both outcomes

Model agreement
disagree
imaging above, clinical history below their cohort medians

Reliability card
few cases
1 of 9 such scans was progression

5 Report
Qwen3.8-27B on vLLM, local

Record for the scan
every value with its source

Local language model
one GPU on site; sees no images, does no arithmetic

Deterministic checker
every number and flag must be in the record, or the text is rejected

PRISM surveillance report SYNTHETIC EXAMPLE

Imaging	0.64
Combined	0.33
Clinical history	0.21

Enhancing tumour is 2.0 cc larger than at the previous scan. The two models disagree, and the 80% conformal set holds both outcomes. This situation has few cases (1 of 9 scans was progression).

B How it was evaluated

MU-Glioma-Post
public development cohort, 5 × 5 cross validation by patient

→ **Models locked**
fitted once on MU; plans committed

→ **LUMIERE, UCSF**
applied unchanged

→ **406**
reports checked